UChicago REU Notes Week 2

Zih-Yu Hsieh

July 12, 2026

Contents

1 Week 2

1.1 Intro

For this portion, I want to start with some basic definitions covered by professor May in his office, about the definition of spectra / prespectra / generalized cohomology theory. I'll probably skip some of the details in the intro to make sure I can keep on track. (Also, I probably involve quite a lot of errors in between...I'll try to fix them)

1.1.1 Brown Representability Theorem, and Generalized Cohomology

Let $h\mathcal{C}$ denotes the homotopy category of based spaces (that admits a CW structure, as on them weak equivalences and homotopy equivalence are the same), then a functor $F : h\mathcal{C}^{\mathrm{op}} \to \mathsf{Ab}$ is *representable*, if there exists a based space $Y \in h\mathcal{C}$, such that $F \cong [\_, Y]$.

Example 1.1. The *reduced cohomology functor* $\tilde{H}^n(\_; A)$ is one, for all $n \geq 0$ and a fixed abelian group A (as coefficient). Let $K(A, n)$ be the *Eilenberg-Maclane Space* of abelian group A and index $n \geq 0$ (i.e. the nth homotopy group $\pi_n(K(A, n)) = A$), then the following group isomorphism holds:

$$\tilde{H}^n(X; A) \cong [X, K(A, n)]$$

Using more adjunction property, since $\pi_n(A, n) = A = \pi_{n+1}(K(A, n+1)) = [S^{n+1}, K(A, n+1)] = [S^n \wedge S^1, K(A, n+1)] \cong [S^n, \Omega K(A, n+1)] = \pi_n(\Omega K(A, n+1))$, there is a homotopy equivalence $K(A, n) \xrightarrow{\simeq} \Omega K(A, n+1)$.

(Yeah, need to understand more about the details)

So, the regular cohomology we see on spaces, are one type of representable functors.

The classification of all such functors is given by *Brown Representability Theorem*:

Theorem 1.2. *Suppose the functor* $F : h\mathcal{C}^{\text{op}} \to \mathsf{Set}$ *satisfies:*

- $F(\bigvee_\alpha X_\alpha) \cong \prod_\alpha F(X_\alpha)$ *(i.e. maps coproduct to product).*
- *F maps* homotopy pushouts *(I believe this is the reduced mapping cylinder?) to* weak pullbacks *(omit the uniqueness of maps). This condition is equivalent to* Mayer-Vietoris axiom *(which seems like a weaker form of the gluing axiom in sheaf theory, and restrict to CW subcomplexes instead of open sets).*

Then, F is representable by some CW complex C, or $F \cong [\_, C]$.

The converse for the theorem also holds, as $[\_, C]$ also satisfies the above axioms (need more details here).

Here is a definition of generalized cohomology:

Definition 1.3. A *generalized cohomology*, is a sequence of spaces $\{E_n\}_{n\geq 0}$, with homotopy equivalences $\{\epsilon_n : E_n \xrightarrow{\simeq} \Omega E_{n+1}\}_{n\geq 0}$.

In a more explicit form, $E^n(X) := [X, E_n]$ has all $n \geq 0$ satisfies $E^n(X) = [X, E_n] \cong [X, \Omega E_{n+1}]$, which is naturally a group (and an abelian group for $n \geq 2$, due to the abelian group structure on $[X, \Omega^m Y]$ for $m \geq 2$).

I remembered the motivation is that, by Brown Representability Theorem, if a sequence of representable functors $E^* : h\mathcal{C}^{\text{op}} \to \mathsf{Set}$ satisfy similar properties to ordinary cohomology (for instance, the connecting morphism $E^n \to E^{n+1}$, and excision axiom?), then Brown Representability Theorem guarantees it must be the above form. So, they're precisely all the generalized cohomology.

1.1.2 Ω-Prespectra, Prespectra, and Spectra

When talking about generalized cohomology, we have a sequence of based spaces $\{E_n\}_{n\geq 0}$, together with homotopy equivalences $\epsilon_n : E_n \xrightarrow{\simeq} \Omega E_{n+1}$, which is a prototype of what's called Ω-*Prespectra*.

In general, fixing a based space $X \in h\mathcal{C}$, the sequence $\{\Sigma^n X\}_{n\geq 0}$ has a collection of natural maps $\{\text{id}_n : \Sigma(\Sigma^n X) \to \Sigma^{n+1} X\}_{n\geq 0}$, which the adjunction (Σ, Ω) guarantees a collection of maps $\{\eta_n : \Sigma^n X \to \Omega(\Sigma^{n+1} X)\}_{n\geq 0}$. In general these maps are not homotopy equivalence, so these type of sequences are a type of *Prespectra*, called *suspension prespectra*. More generally, *Prespectra* is a sequence of spaces $\{X_n\}_{n\geq 0}$, with morphisms $\{\mu_n : X_n \to \Omega X_{n+1}\}_{n\geq 0}$.

For the strongest form possible, if such map $\epsilon_n : E_n \to \Omega E_{n+1}$ is a homeomorphism, then these are called *Spectra*.

Professor May restricts to three specific categories:

- Let $\mathcal{P}$ denotes the *category of prespectra*.
- Let $\mathcal{Q}$ denotes the *category of inclusion prespectra*, meaningeach defining morphism $\mu_n : X_n \to \Omega X_{n+1}$ is an inclusion.
- Let $\mathcal{S}$ denotes the *category of spectra*.

For each category, we require the morphism $\phi : X. \to T.$ to be a sequence of based maps $\phi_n : X_n \to T_n$, such that the following diagram commutes:

$$\begin{tikzcd} \Sigma X_n \arrow[r, "\Sigma\phi_n"] \arrow[d, "\mu'_{X_n}"'] & \Sigma T_n \arrow[d, "\mu'_{T_n}"] \\ X_{n+1} \arrow[r, "\phi_{n+1}"'] & T_{n+1} \end{tikzcd}$$

where the map $\Sigma X_n \to X_{n+1}$ is based on the adjoint property $[\Sigma X_n, X_{n+1}] \cong [X_n, \Omega X_{n+1}]$. Here professor May had talked about how the prototype of the morphism only requires it to commute up to homotopy, but turns out it's too weak as a condition.

With these morphisms, there is an obvious forgetful functor $\mathcal{S} \to \mathcal{Q} \to \mathcal{P}$. It seems to satisfy all the conditions for Freyd's Adjoint Functor Theorem to hold, both of the forgetful functors have a left adjoint.

Here, the left adjoint of $\mathcal{Q} \to \mathcal{P}$ is hard to construct, yet there is a clear construction for the left adjoint of $\mathcal{S} \to \mathcal{Q}$, denoted as $L : \mathcal{Q} \to \mathcal{S}$, by $T. \mapsto (LT).$, where $(LT)_n := \operatorname{colim}_q \Omega^{n+q} T_q$ (due to the inclusion $T_q \to \Omega T_{q+1}$, it induces $\Omega^{n+q} T_q \to \Omega^{n+(q+1)} T_{q+1}$, hence having a sequence of direct system).

Professor May had claimed the homeoomorphism $(LT)_n \cong \Omega(LT)_{n+1}$ due to the fact that Ω commutes with colimit (I do need to work out the detail, as normally Ω as a right adjoint of Σ would be thought of preserving limits instead of colimits).

1.2 Review: Adjoint Functors

This part is converging to the preparation of the theory of monads :) (as professor May had mentioned it quite often as an essential tool to understand operad, so I decided to do related review). The main reference is Riehl's *Category Theory in Context*.

Definition 1.4. Let $\mathcal{C}, \mathcal{D}$ be two categories. Two opposite functors $F : \mathcal{C} \to \mathcal{D}$, $\mathcal{C} \leftarrow \mathcal{D} : G$ are *adjoint pairs* - in particular, F is a *left adjoint of* G,and G is a *right adjoint of* F - if there is a natural isomorphism

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

For any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Notation wise, it's denoted as (F, G) or $F \dashv G$.

More formally, it's a natural isomorphism $(\_)^\sharp : \mathrm{Hom}_{\mathcal{D}}(F(\_), \_) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(\_, G(\_))$ of bifunctors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathsf{Set}$ (here, we also denote $(\_)^\flat$ as the inverse transformation).

This means, given morphism $f : X \to X'$ in $\mathcal{C}$, and morphism $g : Y \to Y'$ in $\mathcal{D}$, we have diagrams as follow:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{D}}(F(X'), Y) \arrow[r, "\sim"] \arrow[d, "_\circ Ff"'] & \mathrm{Hom}_{\mathcal{C}}(X', G(Y)) \arrow[d, "_\circ f"] \\
\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathcal{C}}(X, G(Y))
\end{tikzcd}, \qquad
\begin{tikzcd}
\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \arrow[r, "\sim"] \arrow[d, "g\circ_"'] & \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) \arrow[d, "Gg\circ_"] \\
\mathrm{Hom}_{\mathcal{D}}(F(X), Y') \arrow[r, "\sim"] & \mathrm{Hom}_{\mathcal{C}}(F(X), Y')
\end{tikzcd}$$

The first diagram insists that any morphism $h : F(X') \to Y$ in $\mathcal{D}$ satisfies $(h \circ Ff)^\sharp = h^\sharp \circ f$ in $\mathcal{C}$. The second diagram insists that any morhpism $\ell : F(X) \to Y$ satisfies $(g \circ \ell)^\sharp = Gg \circ \ell^\sharp$. Or, the following diagram holds in $\mathcal{C}$:

$$\begin{tikzcd}
X' \arrow[d, "f"'] \arrow[dr, "(h\circ f)^\sharp"] & \\
X \arrow[r, "h^\sharp"'] & G(Y)
\end{tikzcd}, \qquad
\begin{tikzcd}
X \arrow[r, "\ell^\sharp"] \arrow[dr, "(g\circ\ell)^\sharp"'] & G(Y) \arrow[d, "Gg"] \\
& G(Y')
\end{tikzcd}$$

If use the inverse transformation $(\_)^\flat$, the above can also be phrased as: Any morphism $\tilde{h} : X' \to G(Y)$ satisfies $(\tilde{h} \circ f)^\flat = (\tilde{h})^\flat \circ Ff$, and any morhpism $\tilde{\ell} : X \to G(Y)$ satisfies $(Gg \circ \tilde{\ell})^\flat = g \circ (\tilde{\ell})^\flat$ in $\mathcal{D}$.

A property of adjoint functor is it connects the commutativity of certain diagrams between the two categories:

Lemma 1.5. *Fix arbitrary $f : X \to X'$ in $\mathcal{C}$, $g : Y \to Y'$ in $\mathcal{D}$, and two morphsisms $h : F(X) \to Y$, $h' : F(X') \to Y'$ in $\mathcal{D}$. The commutativity of following two diagrams are equivalent:*

$$\begin{tikzcd}
F(X) \arrow[r, "h"] \arrow[d, "Ff"'] & Y \arrow[d, "g"] \\
F(X') \arrow[r, "h'"'] & Y'
\end{tikzcd} \quad \textit{in } \mathcal{D}, \qquad
\begin{tikzcd}
X \arrow[r, "h^\sharp"] \arrow[d, "f"'] & G(Y) \arrow[d, "Gg"] \\
X' \arrow[r, "h'^\sharp"'] & G(Y')
\end{tikzcd} \quad \textit{in } \mathcal{C}$$

Proof. As a remark, $(\_)^\sharp$ is a bijection on each homset, so two morphisms $F(X) \to Y'$ are equal iff after applying $(\_)^\sharp$ they're equal as morphisms $X \to G(Y')$.

Using the notation from above, we see the following equivalence:

$$g \circ h = h' \circ Ff \iff (g \circ h)^\sharp = (h' \circ Ff)^\sharp \iff Gg \circ h^\sharp = h'^\sharp \circ f$$

Where, the second equivalence is the property from above. The left and right most equalities represent the two commutative diagrams respectively, hence showing the equivalence of commutativity. $\square$

Given the adjoint properties, we have the following two natural isomorphisms, given any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$:

$$(\_)^\sharp : \mathrm{Hom}_\mathcal{D}(F(X), F(X)) \xrightarrow{\sim} \mathrm{Hom}_\mathcal{C}(X, GF(X)), \qquad \mathrm{Hom}_\mathcal{D}(FG(Y), Y) \xleftarrow{\sim} \mathrm{Hom}_\mathcal{C}(G(Y), G(Y)) : (\_)^\flat$$

Hence, we can define a collection of maps $\eta_X := (\mathrm{id}_{F(X)})^\sharp : X \to GF(X)$, and $\epsilon_Y := (\mathrm{id}_{G(Y)})^\flat : FG(Y) \to Y$.

Proposition 1.6. *The two collections form natural transformations $\eta : \mathrm{Id}_\mathcal{C} \to GF$, and $\epsilon : FG \to \mathrm{Id}_\mathcal{D}$.*

Proof. Here, fix arbitrary $f : X \to X'$ in $\mathcal{C}$, and $g : Y \to Y'$ in $\mathcal{D}$. Using **Lemma 1.5**, the following two diagrams have equivalent commutativity:

$$\begin{tikzcd} F(X) \arrow[r, "\mathrm{id}_{F(X)}"] \arrow[d, "Ff"'] & F(X) \arrow[d, "Ff"] \\ F(X) \arrow[r, "\mathrm{id}_{F(X')}"'] & F(X') \end{tikzcd} \text{ in } \mathcal{D}, \qquad \begin{tikzcd} X \arrow[r, "(\mathrm{id}_{F(X)})^\sharp = \eta_X"] \arrow[d, "f"'] & GF(X) \arrow[d, "GFf"] \\ X' \arrow[r, "(\mathrm{id}_{F(X')})^\sharp = \eta_{X'}"'] & GF(X') \end{tikzcd} \text{ in } \mathcal{C}$$

Where, the left side obviously commutes, so the right side commutes, showing naturality of η.

Dually, since $(\epsilon_Y)^\sharp = \mathrm{id}_{G(Y)}$ by the inverse property, we also have the equivalence of the following two diagrams:

$$\begin{tikzcd} FG(Y) \arrow[r, "\epsilon_Y"] \arrow[d, "FGg"'] & Y \arrow[d, "g"] \\ FG(Y) \arrow[r, "\epsilon_{Y'}"'] & Y' \end{tikzcd} \text{ in } \mathcal{D}, \qquad \begin{tikzcd} G(Y) \arrow[r, "(\epsilon_Y)^\sharp = \mathrm{id}_{G(Y)}"] \arrow[d, "f"'] & G(Y) \arrow[d, "Gg"] \\ G(Y') \arrow[r, "(\epsilon_{Y'})^\sharp = \mathrm{id}_{G(Y')}"'] & G(Y') \end{tikzcd} \text{ in } \mathcal{C}$$

Where, the right side obviously commutes, so the left side commutes, showing the naturality of ϵ. $\square$

These natural transformations are named as follow:

Definition 1.7. Given the adjoint pairs $F \dashv G$, the natural transformation $\eta : \mathrm{Id}_\mathcal{C} \to GF$ is called the *unit transformation*, while $\epsilon : FG \to \mathrm{Id}_\mathcal{D}$ is called the *counit transformation.*

They satisfy what's called the *Triangle Identity* as follow:

Proposition 1.8. *The following two diagrams of natural transformation holds:*

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, "\mathrm{id}_F"'] & FGF \arrow[d, "\epsilon_F"] \\ & F \end{tikzcd} \textit{ in } \mathcal{D}^\mathcal{C}, \qquad \begin{tikzcd} G \arrow[r, "\eta_G"] \arrow[dr, "\mathrm{id}_G"'] & GFG \arrow[d, "G\epsilon"] \\ & F \end{tikzcd} \textit{ in } \mathcal{C}^\mathcal{D}$$

Here, $\mathcal{D}^\mathcal{C}$ denotes the *functor categories* of all functors $\mathcal{C} \to \mathcal{D}$, with morphisms being natural transformations between functors.

Proof. Given arbitrary $X \in \mathcal{C}$, the goal is to show the horizontal morphisms of the following diagram are identities:

$$F(X) \xrightarrow{F\eta_X} FGF(X) \xrightarrow{\epsilon_{F(X)}} F(X)$$

Using out notation and the naturality, we see the following equality of morphisms after applying $(\_)^\sharp$:

$$(\epsilon_{F(X)} \circ F\eta_X)^\sharp = (\epsilon_{F(X)})^\sharp \circ \eta_X = \mathrm{id}_{GF(X)} \circ \eta_X = \eta_X : GF(X) \to X$$

But, recall that $\eta_X := (\mathrm{id}_{F(X)})^\sharp$, using the bijectivity of $(\_)^\sharp$, this enforces $\epsilon_{F(X)} \circ F\eta_X = \mathrm{id}_{F(X)}$, showing $\epsilon_F \circ F\eta = \mathrm{id}_F$ as natural transformations.

Dually, given arbitrary $Y \in \mathcal{D}$, the goal is again to show the following morphism is identity:

$$G(Y) \xrightarrow{\eta_{G(Y)}} GFG(Y) \xrightarrow{G\epsilon_Y} G(Y)$$

Similarly, after applying $(\_)^\flat$ we get:

$$(G\epsilon_Y \circ \eta_{G(Y)})^\flat = \epsilon_Y \circ (\eta_{G(Y)})^\flat = \epsilon_Y \circ \mathrm{id}_{FG(Y)} = \epsilon_Y$$

The second equality relies on the fact $(\mathrm{id}_{F(X)})^\sharp = \eta_X$. Which, since $(\_)^\flat$ is again a bijection, and $\epsilon_Y := (\mathrm{id}_G(Y))^\flat$, we see $G\epsilon_Y \circ \eta_{G(Y)} = \mathrm{id}_{G(Y)}$, showing $G\epsilon \circ \eta_G = \mathrm{id}_G$ as natural transformations. □

This will be an important result when classifying monads. We also want to say the converse of Triangle Identity holds, that it generates an adjoint pair:

Lemma 1.9. *Suppose $F : \mathcal{C} \to \mathcal{D}$ and $\mathcal{C} \leftarrow \mathcal{D} : G$ are two opposite functors, together with natural transformations $\eta : \mathrm{Id}_{\mathcal{C}} \to GF$ and $\epsilon : FG \to \mathrm{Id}_{\mathcal{D}}$ such that Triangle Identity holds:*

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, "\mathrm{id}_F"'] & FGF \arrow[d, "\epsilon_F"] & \text{in } \mathcal{D}^{\mathcal{C}}, \\ & F & \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, "\eta_G"] \arrow[dr, "\mathrm{id}_G"'] & GFG \arrow[d, "G\epsilon"] & \text{in } \mathcal{C}^{\mathcal{D}} \\ & F & \end{tikzcd}$$

then $F \dashv G$ is an adjoint pair.

Proof. Given any $X \in \mathcal{C}$ (associated to $\eta_X : X \to GF(X)$) and $Y \in \mathcal{D}$ (associated to $\epsilon_Y : FG(Y) \to Y$), for any morphism $h : F(X) \to Y$, we have a corresponding morphism $Gh : GF(X) \to G(Y)$. This defines the following map:

$$(\_)^\sharp : \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \to \mathrm{Hom}_{\mathcal{C}}(X, G(Y)), \quad h^\sharp := Gh \circ \eta_X$$

Similarly, for any morphism $\ell : X \to G(Y)$, we have a corresponding morphism $F\ell : F(X) \to FG(Y)$. This again defines the following map:

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \leftarrow \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) : (\_)^\flat, \quad \ell^\flat := \epsilon_Y \circ F\ell$$

These are evidently functorial. To show they're mutual inverses, we expand the following:

$$(h^\sharp)^\flat = \epsilon_Y \circ Fh^\sharp = \epsilon_Y \circ F(Gh \circ \eta_X) = \epsilon_Y \circ FGh \circ F\eta_X$$

Expanding the right hand side in terms of diagram (includes the triangle identity and naturality of ϵ), we get:

$$\begin{tikzcd} F(X) \arrow[r, "F\eta_X"] \arrow[rr, bend left, "\mathrm{id}_{F(X)}"] & FGF(X) \arrow[r, "\epsilon_{F(X)}"] \arrow[d, "FGh"'] & F(X) \arrow[d, "h"] \\ & FG(Y) \arrow[r, "\epsilon_Y"'] & Y \end{tikzcd}$$

So, we see $(h^\sharp)^\flat = \epsilon_Y \circ FGh \circ F\eta_X = h$ by the above commutative diagram.

Expand the dual diagram with the other triangle identity and naturality of η, we'll get that $(\ell^\flat)^\sharp = \ell$ instead, showing the two are mutual inverses. $\square$

1.3 Monads and Relations to Adjoint Functors

Let's first observe the case when we have an adjoint pair $F \dashv G$ from above, inducing an endofunctor $GF : \mathcal{C} \to \mathcal{C}$, with unit transformation $\eta : \mathrm{Id}_{\mathcal{C}} \to GF$, together with *multiplication transformation* $\mu := G\epsilon_F : GFGF \to GF$ (as we have $\epsilon : FG \to \mathrm{Id}_{\mathcal{D}}$). Notice they provide the following identities:

- This can be viewed as *Associativity of Multiplication*:

$$\begin{tikzcd} (GF)^3 \arrow[r, "GF\mu"] \arrow[d, "\mu_{GF}"'] & (GF)^2 \arrow[d, "\mu"] \\ (GF)^2 \arrow[r, "\mu"'] & GF \end{tikzcd}$$

  Fix arbitrary $X \in \mathcal{C}$, expand the definition, we get the following diagram:

$$\begin{tikzcd} GFG(FGF(X)) \arrow[r, "GFG\epsilon_{F(X)}"] \arrow[d, "G\epsilon_{FGF(X)}"'] & GFG(F(X)) \arrow[d, "G\epsilon_{F(X)}"] \\ G(FGF(X)) \arrow[r, "G\epsilon_{F(X)}"'] & G(F(X)) \end{tikzcd}$$

  Disregard the left most G, the above diagram commutes because $\epsilon : FG \to \mathrm{Id}_{\mathcal{D}}$ is a natural transformation.

- This can be viewed as *Unit Property*:

$$\begin{tikzcd} GF \arrow[r, "\eta_{GF}"] \arrow[dr, "\mathrm{id}_{GF}"'] & (GF)^2 \arrow[d, "\mu"] & GF \arrow[l, "GF\eta"'] \arrow[dl, "\mathrm{id}_{GF}"] \\ & GF & \end{tikzcd}$$

  Using the definition $\mu := G\epsilon_F$, we have the following two equalities using *Triangle Identities of Adjunction*:

$$G\epsilon_F \circ \eta_{GF} = (G\psi_F \circ \eta_G)_F = (\mathrm{id}_G)_F = \mathrm{id}_{GF}$$

$$G\epsilon_F \circ GF\eta = G(\epsilon_F \circ F\eta) = G(\mathrm{id}_F) = \mathrm{id}_{GF}$$

  Thse two identities reflect the commutativity of the above diagram.

This reflects the structure of *Monad*:

Definition 1.10. Given a category $\mathcal{C}$, a *Monad* on $\mathcal{C}$ consists of the following:

- An endofunctor $T : \mathcal{C} \to \mathcal{C}$.
- A *unit transformation* $\eta : \mathrm{Id}_{\mathcal{C}} \to T$.
- A *multiplication transformation* $\mu : T^2 \to T$.

There are extra requirements of satisfying the following two diagrams as endofunctors on $\mathcal{C}$:

$$\begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu_T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd} , \qquad \begin{tikzcd} T \arrow[r, "\eta_T"] \arrow[dr, "\mathrm{id}_T"'] & T^2 \arrow[d, "\mu"] & T \arrow[l, "T\eta"'] \arrow[dl, "\mathrm{id}_T"] \\ & T & \end{tikzcd}$$

Which are analogs to *associativity* and *unit* of monoids.

Which, starting portion shows that any adjoint pair $(F \dashv G)$ generates a monad on category $\mathcal{C}$ (the source category of the left adjoint F).

Example 1.11. As a more relevant example, let $\mathcal{T}^*$ denotes the category of based spaces, $\mathcal{T}$ denotes the category of topological spaces, there is a natural forgetful functor $F : \mathcal{T}^* \to \mathcal{T}$ that forgets the based spaces.

It has a left adjoint $(\_)_* : \mathcal{T} \to \mathcal{T}^*$, where $(X)_* := X \sqcup \{*\}$ by adding a point as basepoint, and any map $f : X \to Y$ naturally extends to a based map $\tilde{f} : X_* \to Y_*$ where $\tilde{f}|_X = f$, and $\tilde{f}(*) = *$.

For any space $X \in \mathcal{T}$ and based space $Y \in \mathcal{T}^*$ (with $y_0 \in Y$ as basepoint), the natural isomorphism $\mathrm{Hom}_{\mathcal{T}^*}(X_*, Y) \cong \mathrm{Hom}_{\mathcal{T}}(X, Y)$ is given by $g \mapsto g|_X$, and inverse is $f \mapsto \tilde{f}$.

In this case, the unit map $\eta_X = \mathrm{id}_{X_*}|_X$, which becomes the canonical inclusion $X \hookrightarrow X_*$; the counit map $\epsilon_Y = \tilde{\mathrm{id}_Y}$, which is the map $Y_* \to Y$ that collapse the new basepoint $*$ to the original basepoint y_0. So, the multiplication map $\eta_X = F\epsilon_{X_*} : (X_*)_* \to X_*$ by collapsing the two added basepoints (in different iteration) to the unique basepoint (with one iteration).

Since collapsing of three added points to one doesn't care about the collapse order, this explicitly forms a monad (or using the adjunction property this directly follows :)

Another question to ask: Is every monad of a category induced by adjoint pairs of functors? The answer is yes, but one needs to introduce some relevant categories constructed out of the monads.

Definition 1.12. Let (T, η, μ) be a monad on category $\mathcal{C}$. Its *Eilenberg-Moore category* for T (or the *category of T-algebras*), is a category $\mathcal{C}^T$ consists of:

- Objects of pairs (A, a), where $A \in \mathcal{C}$ is an object, and $a : T(A) \to A$ is a morphism, so the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[dr, "\mathrm{id}_A"'] & T(A) \arrow[d, "a"] \\ & A \end{tikzcd}, \qquad \begin{tikzcd} T^2(A) \arrow[r, "\mu_A"] \arrow[d, "Ta"'] & T(A) \arrow[d, "a"] \\ T(A) \arrow[r, "a"'] & A \end{tikzcd}$$

- A morphism $f : (A, a) \to (B, b)$ is a morhpism $f : A \to B$, such that the following diagram commutes:

$$\begin{tikzcd} T(A) \arrow[r, "Tf"] \arrow[d, "a"'] & T(B) \arrow[d, "b"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

I think intuitively $T(A)$ consists of data about "pairs of actions", while a encodes how this action appears on A. The square provides the "associativity" of action (for instance, $(ab)\dot{v} = a(b\dot{v})$ as action on modules). I do need to agree with professory May, that this system is more like a module instead of algebra in the traditional sense.

For this category, there is an obvious "forgetful functor" $G^T : \mathcal{C}^T \to \mathcal{C}$ by $G^T(A, a) = A$ and $G^T f = f$ for any $f : (A, a) \to (B, b)$.

Similarly, since $\mathcal{C}$ is given a monad, there is also an obvious "algebra functor" $F^T : \mathcal{C} \to \mathcal{C}^T$, by $F^T(A) := (T(A), \mu_A)$, and $F^T f = Tf$ for any $f : A \to B$. This is because μ_A satisfies the following two identities, by the definition of monad:

$$\begin{tikzcd} T(A) \arrow[r, "\eta_{T(A)}"] \arrow[dr, "\mathrm{id}_{T(A)}"'] & T^2(A) \arrow[d, "\mu_{T(A)}"] \\ & T(A) \end{tikzcd} \quad , \qquad \begin{tikzcd} T^2(T(A)) \arrow[r, "\mu_{T(A)}"] \arrow[d, "T\mu_A"'] & T(T(A)) \arrow[d, "\mu_A"] \\ T(T(A)) \arrow[r, "\mu_A"'] & T(A) \end{tikzcd}$$

Also, since $\mu : T^2 \to T$ is a natural transformation, the following diagram commutes:

$$\begin{tikzcd} T(T(A)) \arrow[r, "Tf"] \arrow[d, "\mu_A"'] & T(T(B)) \arrow[d, "\mu_B"] \\ T(A) \arrow[r, "f"'] & T(B) \end{tikzcd}$$

showing $Tf : (T(A), \mu_A) \to (T(B), \mu_B)$ is a morphism.

Here comes an important result regarding this category:

Theorem 1.13. *The two functors are adjoint pairs ($F^T \dashv G^T$), and the induced monad is (T, η, μ).*

Proof. Given arbitrary $A \in \mathcal{C}$ and $(B, b) \in \mathcal{C}^T$, we shall show a natural isomorphism between the two:

$$\Phi : \mathrm{Hom}_{\mathcal{C}^T}((T(A), \mu_A), (B, b)) \rightleftarrows \mathrm{Hom}_{\mathcal{C}}(A, B) : \Psi$$

Given a morphism $f : (T(A), \mu_A) \to (B, b)$, recall the existence of unit morphism $\eta_A : A \to T(A)$, so $\Phi(f) := f \circ \eta_A : A \to T(A) \to B$ defines a map.

Conversely, given any morhpism $g : A \to B$, it naturally promotes to $Tg : (T(A), \mu_A) \to (T(B), \mu_B)$; and, since $(B, b) \in \mathcal{C}^T$, postcomposing with the morphism $b : (T(B), \mu_B) \to (B, b)$ provides $b \circ Tg : (T(A), \mu_A) \to (T(B), \mu_B) \to (B, b)$. Define $\Psi(g) := b \circ Tg$.

Remark 1.14. The reason why $b : T(B) \to B$ is also a morphism in $\mathcal{C}^T$, is because of the commutative square as follow by the fact $(B, b) \in \mathcal{C}^T$:

$$\begin{tikzcd} T(T(B)) \arrow[r, "\mu_B"] \arrow[d, "Tb"'] & T(B) \arrow[d, "b"] \\ T(B) \arrow[r, "b"'] & B \end{tikzcd} \quad \Longrightarrow \quad \begin{tikzcd} T(T(B)) \arrow[r, "Tb"] \arrow[d, "\mu_B"'] & T(B) \arrow[d, "b"] \\ T(B) \arrow[r, "b"'] & B \end{tikzcd}$$

The right hand side is the transpose of the left hand side, also the defining diagram of $b : (T(B), \mu_B) \to (B, b)$ as a morphism in $\mathcal{C}^T$.

The two maps Φ, Ψ are for sure functorial, hence suffices to check they're mutual inverses. Let's compute them respectively:

- $$\Psi(\Phi(f)) = \Psi(f \circ \eta_A) = b \circ T(f \circ \eta_A) = b \circ Tf \circ T\eta_A$$

  To reduce this, consider the following diagram, using triangle identities of monad and the assumptions about morphisms:

  $$\begin{tikzcd} T(A) \arrow[r, "T\eta_A"] \arrow[dr, "\mathrm{id}_{T(A)}"'] & T^2(A) \arrow[r, "Tf"] \arrow[d, "\mu_A"'] & T(B) \arrow[d, "b"] \\ & T(A) \arrow[r, "f"'] & B \end{tikzcd}$$

So, we see $b \circ Tf \circ T\eta_A = f \circ \mathrm{id}_{T(A)} = f : T(A) \to B$ as morphism in $\mathcal{C}$ (and hence in $\mathcal{C}^T$), showing $\Psi(\Phi(f)) = f$.

- $$\Phi(\Psi(g)) = \Phi(b \circ Tg) = b \circ Tg \circ \eta_A$$

  To reduce this, again consider similar definitions (and the fact $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is a natural transformation):

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "\eta_A"'] & B \arrow[d, "\eta_B"'] \arrow[dr, "\mathrm{id}_B"] & \\ T(A) \arrow[r, "Tg"'] & T(B) \arrow[r, "b"'] & B \end{tikzcd}$$

  So, we get $b \circ Tg \circ \eta_A = \mathrm{id}_B \circ g = g$ as morphism in $\mathcal{C}$, hence $\Phi(\Psi(g)) = g$.

This shows $(F^T \dashv G^T)$ in fact forms an adjoint pair.

To show they generate monad (T, η, μ) on $\mathcal{C}$, notice the following action on each object $A \in \mathcal{C}$ / morphism $f : A \to B$:

$$G^T F^T(A) = G^T(T(A), \mu_A) = T(A), \quad G^T F^T f = G^T(Tf) = Tf$$

Hence, $G^T F^T = T$ as endofunctor on $\mathcal{C}$.

Similarly, the unit transformation $\eta' : \mathrm{Id}_{\mathcal{C}} \to G^T F^T$ as $\eta'_A := \Phi(\mathrm{id}_{(T(A),\mu_A)}) = \mathrm{id}_{T(A)} \circ \eta_A = \eta_A$, showing $\eta' = \eta$.

Finally, we have counit transformation $\epsilon' : F^T G^T \to \mathrm{Id}_{\mathcal{C}^T}$ as $\epsilon'_{(B,b)} := \Psi(\mathrm{id}_B) = b \circ T(\mathrm{id}_B) = b \circ \mathrm{id}_{T(B)} = b$. So, the induced multiplication transformation $\mu' := G^T \epsilon'_{F^T} : (G^T F^T)^2 \to G^T F^T$ satisfies $\mu'_A = G^T \epsilon'_{(T(A),\mu_A)} = G^T(\mu_A) = \mu_A$, showing $\mu' = \mu$.

Hence, $(F^T \dashv G^T)$ induces monad (T, η, μ). $\square$

This demonstrates each monad is induced by some adjoint pairs.

Example 1.15. For the basepoint monad demonstrated in **Example 1.11**, the Eilenber-Moore category $\mathcal{T}^{(\text{-})_*}$ consists of pairs (X, a) where $a : X_* \to X$ is a map that satisfies the following:

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "\mathrm{id}_X"'] & X_* \arrow[d, "a"] \\ & X \end{tikzcd}, \qquad \begin{tikzcd} (X_*)_* \arrow[r, "Ta"] \arrow[d, "\mu_X"'] & X_* \arrow[d, "a"] \\ X_* \arrow[r, "a"'] & X \end{tikzcd}$$

The first diagram enforces $a|_X = \mathrm{id}_X$ (as $\eta_X : X \to X_*$ is the canonical inclusion $X \hookrightarrow X \sqcup \{*\}$). Given this data, the second diagram automatically commutes (by checking where the added points are sent to), so we see a is purely determined by $a(*) \in X$, in other words, it's specified by (X, x_0) for $x_0 \in X$.

Then, given a morphism $f : (X, a) \to (Y, b)$, the following diagram commutes:

$$\begin{tikzcd} X_* \arrow[r, "\tilde{f}"] \arrow[d, "a"'] & Y_* \arrow[d, "b"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

Which, $f(a(*)) = b(\tilde{(f)}(*)) = b(*)$ is enforced, showing f must map the chosen point in X to the chosen point in Y.

Hence, the functor $\mathcal{T}^{(-)_*} \to \mathcal{T}^*$ by $(X, a) \mapsto (X, a(x))$, and $f \mapsto f$ is in fact an equivalence of category. So, the Eilenberg-Moore category of the basepoint functor is the category of based space.

Afterward, there's more question to ask: Given a monad on $\mathcal{C}$, how many ways are there to create adjoint pairs which induce the provided monad? Can we find such universal adjoint pair? For this, there are more definitions to make:

Definition 1.16. Given a monad (T, η, μ) on category $\mathcal{C}$. The *Kleisli category* $\mathcal{C}_T$ consists of:

- Objects are carried over from C.
- Morphism $f : A \rightsquigarrow B$ in $\mathcal{C}_T$ is a morphism $f : A \to T(B)$ in $\mathcal{C}$. Under this definition, $\eta_A : A \rightsquigarrow A$ serve as the identity morphism of $A \in \mathcal{C}_T$, while given $f : A \rightsquigarrow B$ and $g : B \rightsquigarrow C$, the following morphism in $\mathcal{C}$ serves as the composition of the two:

$$g \circ_* f := A \xrightarrow{f} T(B) \xrightarrow{Tg} T(T(C)) \xrightarrow{\mu_C} T(C)$$

Proof. First, let's show associativity: Given $f : A \rightsquigarrow B$, $g : B \rightsquigarrow C$, and $h : C \rightsquigarrow D$, the two ways of composition provide the following in $\mathcal{C}$:

$$h \circ_* (g \circ_* f) = A \xrightarrow{f} T(B) \xrightarrow{Tg} T^2(C) \xrightarrow{\mu_C} T(C) \xrightarrow{Th} T^2(D) \xrightarrow{\mu_D} T(D)$$

$$(h \circ_* g) \circ_* f = A \xrightarrow{f} T(B) \xrightarrow{Tg} T^2(C) \xrightarrow{T^2h} T^3(D) \xrightarrow{T\mu_D} T^2(D) \xrightarrow{\mu_D} T(D)$$

To show the two procedures agree, it suffices to show the middle right portion $\mu_D \circ Th \circ \mu_C = \mu_D \circ T\mu_D \circ T^2h$. However, this follows by the monad property:

$$\begin{tikzcd}
T^2(C) \arrow[r, "T^2h"] \arrow[d, "\mu_C"'] & T^3(D) \arrow[r, "T\mu_D"] \arrow[d, "\mu_{T(D)}"] & T^2(D) \arrow[d, "\mu_D"] \\
T(C) \arrow[r, "Th"'] & T^2(D) \arrow[r, "\mu_D"'] & T(D)
\end{tikzcd}$$

the left square commutes by naturality of μ, the right square commutes by associativity of μ.

To show η_A serves as identity, we compute the following given $f : A \rightsquigarrow B$ and $g : Z \rightsquigarrow A$:

- $$f \circ_* \eta_A = A \xrightarrow{\eta_A} T(A) \xrightarrow{Tf} T^2(B) \xrightarrow{\mu_B} T(B)$$

  This can be simplified using triangle identity, and the fact $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is a natural transformation:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "\eta_A"'] & T(B) \arrow[d, "\eta_{T(B)}"'] \arrow[dr, "\mathrm{id}_{T(B)}"] & \\
T(A) \arrow[r, "Tf"'] & T^2(B) \arrow[r, "\mu_B"'] & T(B)
\end{tikzcd}$$

  So, we deduce $f \circ_* \eta_A = \mu_B \circ Tf \circ \eta_A = \mathrm{id}_{T(B)} \circ f = f$.

- $$\eta_A \circ_* g = Z \xrightarrow{g} T(A) \xrightarrow{T\eta_A} T^2(A) \xrightarrow{\mu_A} T(A)$$

  Notice that $\mu_A \circ T\eta_A = \mathrm{id}_{T(A)}$ in $\mathcal{C}$ (by triangle identity),so we see $\mu_A \circ_* T\eta_A = \mu_A \circ T\eta_A \circ g = \mathrm{id}_{T(A)} \circ g = g$.

$\square$

For this category, there is a functor $G_T : \mathcal{C}_T \to \mathcal{C}$, by $G_T(A) = T(A)$, and $G_T f := \mu_B \circ Tf : T(A) \to T^2(B) \to T(B)$ for arbitrary $A \in \mathcal{C}_T$ and morphism $f : A \rightsquigarrow B$. This is functorial, as $G_T \eta_A = \mu_A \circ T\eta_A = \mathrm{id}_{T(A)}$ by applying triangle identity, and given $f : A \rightsquigarrow B$, $g : B \rightsquigarrow C$, we have the following diagram commutes:

$$\begin{tikzcd}
T(A) \arrow[r, "Tf"] & T^2(B) \arrow[r, "T^2 g"] \arrow[d, "\mu_B"'] & T^3(C) \arrow[r, "T\mu_C"] \arrow[d, "\mu_{T(C)}"'] & T^2(C) \arrow[d, "\mu_C"] \\
& T(B) \arrow[r, "Tg"'] & T^2(C) \arrow[r, "\mu_C"'] & T(C)
\end{tikzcd}$$

This shows $G_T(g \circ_* f) = \mu_C \circ T(\mu_C \circ Tg \circ f) = \mu_C \circ T\mu_C \circ T^2 g \circ Tf = (\mu_C \circ Tg) \circ (\mu_B \circ Tf) = G_T g \circ G_T f$.

Conversely, there is a "coalgebra functor" $F_T : \mathcal{C} \to \mathcal{C}_T$, by $F_T(A) = A$, and $F_T f := \eta_B \circ f : A \to B \to T(B)$ for all $A \in \mathcal{C}$ and $f : A \to B$ in $\mathcal{C}$. This is again functorial, as $F_T \mathrm{id}_A := \eta_A \circ \mathrm{id}_A = \eta_A$ is the identity in $\mathcal{C}_T$, and given $f : A \to B$, $g : B \to C$ the following diagram commutes:

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[dr, "F_T f"'] & B \arrow[r, "g"] \arrow[d, "\eta_B"'] & C \arrow[d, "\eta_C"] & \\
& T(B) \arrow[r, "Tg"] \arrow[dr, "T(F_T g)"'] & T(C) \arrow[d, "T\eta_C"] \arrow[dr, "\mathrm{id}_{T(C)}"] & \\
& & T^2(C) \arrow[r, "\mu_C"'] & T(C)
\end{tikzcd}$$

(Two left triangles are definition of F_T, the square is the naturality of $\eta : \mathrm{Id}_C \to T$, and bottom right is the triangle identity of monad).

This shows $F_T(g \circ f) = \eta_C \circ (g \circ f) = \mathrm{id}_{T(C)} \circ \eta_C \circ g \circ f = \mu_C \circ T(F_T g) \circ F_T f = F_T g \circ_* F_T f$.

Similar to the Eilenberg-Moore category, we have the adjunction:

Theorem 1.17. *The two functors are adjoint pairs ($F_T \dashv G_T$), and the induced monad is (T, η, μ).*

Proof. Similar manner, we ought to show a natural isomorphism between the two homsets:

$$\Phi : \mathrm{Hom}_{\mathcal{C}_T}(A, B) \rightleftarrows \mathrm{Hom}_{\mathcal{C}}(A, T(B)) : \Psi$$

In fact, by definition these two sets are equal, as a morphism $f : A \rightsquigarrow B$ in $\mathcal{C}_T$, is precisely a morphism $f : A \to T(B)$, so the "identity map" between the two sets provide an obvious bijection.

About functoriality, this is established by the properties of morphisms that're established beforehand, here we can comfortably omit it :)

It remains to check they do form the monad T: Fix $A \in \mathcal{C}$, and morphism $f : A \to B$, we get the following description:

$$G_T F_T(A) = G_T(A) = T(A), \quad G^T(F^T f) = G_T(\eta_B \circ f) = \mu_B \circ T(\eta_B \circ f) = \mu_B \circ T\eta_B \circ Tf = Tf$$

The morphism equality is due to triangle identity, $\mu_B \circ T\eta_B = \mathrm{id}_{T(B)}$. This shows $T = G_T F_T$.

About the unit transformation $\eta' : \mathrm{Id}_{\mathcal{C}} \to G_T F_T$ is given by $\eta'_A = \Phi(\eta_A) = \eta_A$ (because η_A is the identity of A in $\mathcal{C}_T$), showing $\eta' = \eta$.

About the counit transformation $\epsilon' : F_T G_T \to \mathrm{Id}_{\mathcal{C}_T}$, every $B \in \mathcal{C}_T$ satisfies $\epsilon'_B = \Psi(\mathrm{id}_{T(B)}) = \mathrm{id}_{T(B)} : T(B) \rightsquigarrow B$. Then, the multiplication transformation $\mu' := G_T \epsilon_{F_T}$ satisfies $\mu'_A = G_T \epsilon_{F_T(A)} = G_T(\mathrm{id}_{T(A)}) = \mu_A \circ T\mathrm{id}_{T(A)} = \mu_A \circ \mathrm{id}_{T^2(A)} = \mu_A$, showing $\mu' = \mu$.

So, the given adjunction does induce the monad (T, η, μ), finishing the proof. $\square$

Finally, the universality of these two categories are given as follow:

Theorem 1.18. *In the* adjoint category Adg_T *(categories of all category with adjoint functors between $\mathcal{C}$, inducing monad T; the morphisms are functors that commutes with the left/right adjoints for each category), the Kleisli category $(\mathcal{C}_T, F_T \dashv G_T)$ is initial, while Eilenberg-Moore category $(\mathcal{C}^T, F^T \dashv G^T)$ is final.*

Lemma 1.19. *The canonical functor $\mathcal{C}_T \to \mathcal{C}^T$ is fully faithful, with essential image being all the free T-algebra (all pairs $(T(A), \mu_A)$).*

1.4 Monad Algebra and Monadicity (ND)

Fix a monad (T, η, μ) on category $\mathcal{C}$, we're mainly concerned with the Eilenber-Moore category $\mathcal{C}^T$ (or the category of T-algebras). In particular, we wish to study the free algebras, namely the pair $(T(A), \mu_A) \in \mathcal{C}^T$, and see under what situation this covers all the possible T-algebras.

The first goal is to show in $\mathcal{C}^T$, any T-algebra (A, a) satisfies the following coequalizer diagram:

$$\begin{tikzcd} (T^2(A), \mu_{T(A)}) \arrow[r, shift left, "Ta"] \arrow[r, shift right, "\mu_A"'] & (T(A), \mu_A) \arrow[r, "a"] & A \end{tikzcd}$$

For this, let's do some definition:

Definition 1.20. A *split coequalizer diagram* consists of the following morphisms:

$$\begin{tikzcd} X \arrow[r, shift left, bend left, "f"] \arrow[r, shift right, bend right, "g"'] & Y \arrow[r, "h"] \arrow[l, bend left=60, "t"] & Z \arrow[l, bend left=60, "s"] \end{tikzcd}$$

with the requirement $h \circ f = h \circ g$, $h \circ s = \text{id}_Z$, $g \circ t = \text{id}_Y$, and $f \circ t = s \circ h$ as endomorphism on Y.

Lemma 1.21. *Given a split coequalizer diagram as above, then $h : Y \to Z$ serves as a categorical coequalizer of $f, g : X \to Y$. Also, it's an* absolute colimit*, meaning all functor preserves this coequalizer.*

Proof. It is evident why all functor preserves this, simply because functor preserves identity, hence split coequalizer diagram is preserved under any functor. Hence, it suffices to show a split coequalizer diagram is a coequalizer.

Let $h' : Y \to Z'$ satisfies $h' \circ f = h' \circ g$, consider the morphism $h' \circ s : Z \to Y \to Z'$. Notice its composition with h provides the following:

$$(h' \circ s) \circ h = h' \circ (f \circ t) = (h' \circ g) \circ t = h' \circ \text{id}_Y = h'$$

Hence, the following factorization holds:

$$\begin{tikzcd} X \arrow[r, shift left, bend left, "f"] \arrow[r, shift right, bend right, "g"'] & Y \arrow[r, "h"] \arrow[dr, "h'"'] & Z \arrow[d, "h' \circ s"] \\ & & Z' \end{tikzcd}$$

Then, the uniqueness of $h' \circ s$ is due to the fact h is an epimorphism: Recall that $h \circ s = \text{id}_Z$, with id_Z being epimorphism, so is h. Then, if $\psi : Z \to Z'$ also satisfies $(h' \circ s) \circ h = h' = \psi \circ h$, right cancellative property guarantees $\psi = h' \circ s$. This shows $h : Y \to Z$ is a coequalizer of f, g. $\square$

For instance, given $(A, a) \in \mathcal{C}^T$ a T-algebra, the following diagram is a split coequalizerin $\mathcal{C}$ (**Remark:** NOT in $\mathcal{C}^T$):

$$\begin{tikzcd} T^2(A) \arrow[r, shift left, bend left, "Ta"] \arrow[r, shift right, bend right, "\mu_A"'] & T(A) \arrow[r, "a"] \arrow[l, bend left=60, "\eta_{T(A)}"] & A \arrow[l, bend left=60, "\eta_A"] \end{tikzcd}$$

Since $a \circ Ta = a \circ \mu_A$ using associativity diagram while $a \circ \eta_A = \text{id}_A$ by unit diagram of monad algebra, $\mu_A \circ \eta_{T(A)} = \text{id}_{T(A)}$ by triangle identity of monad, and finally $Ta \circ \eta_{T(A)} = \eta_A \circ a$ due to the fact $\eta : \text{Id}_{\mathcal{C}} \to T$ is a natural transformation.

More general result need more definition:

Definition 1.22. Given a functor $U : \mathcal{D} \to \mathcal{C}$.

- a *U-split coequalizer* of $f, g : X \to Y$ in $\mathcal{D}$, is an object $Z \in \mathcal{C}$, together with morphism $h : U(Y) \to Z$, $s : Z \to U(Y)$, and $t : U(Y) \to U(X)$, such that the following diagram is a split coequalizer in $\mathcal{C}$:

$$\begin{tikzcd} U(X) \arrow[r, shift left, "Uf"] \arrow[r, shift right, "Ug"'] & U(Y) \arrow[r, "h"] \arrow[l, bend left=60, "t"] & Z \arrow[l, bend left=60, "s"] \end{tikzcd}$$

- *U creates coequalizers of U-split pairs*, if any U-split coequalizer admits a coequalizer $h' : Y \to Z'$ in $\mathcal{D}$, such that its image $Uh' : U(Y) \to U(Z')$ is isomorphic to $h : U(Y) \to Z$ in $\mathcal{C}$.

- *U strictly creates coequalizers of U-split pairs*, if it creates coequalizer of U-split pairs, while such lift h' in $\mathcal{D}$ is unique (up to isomorphism in $\mathcal{D}$).

d

1.5 Review: Monoidal and Symmetric Monoidal Categories

This part follows *Operads, Algebras, and Modules* by May. Let's first review on monoidal/symmetric monoidal category:

Definition 1.23. A category $\mathcal{V}$ is a *monoidal caetgory*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ (called *monoidal product*), and an object $1 \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (\_ \otimes \_) \otimes \_ \xrightarrow{\sim} \_ \otimes (\_ \otimes \_)$ of functors $\mathcal{V} \times \mathcal{V} \times \mathcal{V} \xrightarrow{\sim} \mathcal{V}$. (So, $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ in a canoncial way).

- The *left unit* $\lambda : 1 \otimes \_ \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($1 \otimes X \cong X$ in a canonical way).

- The *right unit* $\rho : \_ \otimes 1 \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($X \otimes 1 \cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{tikzcd} (X \otimes 1) \otimes Y \arrow[rr, "a_{X,1,Y}"] \arrow[dr, "\rho_X \otimes \mathrm{id}_Y"'] & & X \otimes (1 \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes \lambda_Y"] \\ & X \otimes Y & \end{tikzcd}$$

- *Pentagon Axiom:*

$$\begin{tikzcd} ((X \otimes Y) \otimes Z) \otimes W \arrow[rr, "a_{X \otimes Y, Z, W}"] \arrow[d, "a_{X,Y,Z} \otimes \mathrm{id}_W"'] & & (X \otimes Y) \otimes (Z \otimes W) \arrow[d, "a_{X,Y,Z \otimes W}"] \\ (X \otimes (Y \otimes Z)) \otimes W \arrow[dr, "a_{X, Y \otimes Z, W}"'] & & X \otimes (Y \otimes (Z \otimes W)) \\ & X \otimes ((Y \otimes Z) \otimes W) \arrow[ur, "\mathrm{id}_X \otimes a_{Y,Z,W}"'] & \end{tikzcd}$$

Based on *Maclane's Coherence Theorem*, the above two axioms guarantees all finite sequences of "associators" (the morphisms involving a, λ, ρ and their inverses only) between finite products agree, namely there is a unique isomorphism $P_1 \to P_2$, where P_i are (possibly) different orders of performing product on $X_1, ..., X_n \in \mathcal{V}$.

This guarantees all monoidal category to be equivalent to a *strict monoidal category*,namely all a, λ, ρ above are "identity transformations", or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 1.24. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : \_{1} \otimes \_{2} \xrightarrow{\sim} \_{2} \otimes \_{1}$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "B_{X,Y} \otimes \mathrm{Id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "B_{X, Y\otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
(Y \otimes X) \otimes Z \arrow[r, "a_{Y,X,Z}"'] & Y \otimes (X \otimes Z) \arrow[r, "\mathrm{id}_Y \otimes B_{X,Z}"'] & Y \otimes (Z \otimes X)
\end{tikzcd}$$

$$\begin{tikzcd}
X \otimes (Y \otimes Z) \arrow[d, "\mathrm{id}_X \otimes B_{Y,Z}"'] & (X \otimes Y) \otimes Z \arrow[l, "a_{X,Y,Z}"'] \arrow[r, "B_{X\otimes Y, Z}"] & Z \otimes (X \otimes Y) \\
X \otimes (Z \otimes Y) & (X \otimes Z) \otimes Y \arrow[l, "a_{X,Z,Y}"] \arrow[r, "B_{X,Z} \otimes \mathrm{id}_Y"'] & (Z \otimes X) \otimes Y \arrow[u, "a_{Z,X,Y}"']
\end{tikzcd}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are guaranteeing the ability to perform 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees isomorphism between all permutations of finite products. The symmetric axiom further imposes that there is a single way of doing such permutation (so we don't have "nontrivial braiding", namely $B_{Y,X} \circ B_{X,Y} : X \otimes Y \to Y \otimes X \to X \otimes Y$ is not identity).

From my naive understanding, the coherence theorem guarantees a treatment $X \otimes Y = Y \otimes X$ without loss of generality.

1.6 Operads and Operadic Algebra

When searching up extra sources, there is another approach using multicategories (having multimorphisms $\mathrm{Hom}(a_1, ..., a_n; b)$, diagramatically like a "fork", and different ways of compiling the forks. I think this understanding is more intuitive than traditional definition).

However for understanding in the literature of homotopy theory, I think I'll focus on the original definition. This part cf. May's *Operads, Algebras, and Modules*, *Geometry of Iterated Loop Spaces*, with supplemnts through an arxiv document (https://arxiv.org/pdf/2508.01886) by Felicia Ferraioli, and Maclane's *Category Theory for Working Mathematicians*.

Definition 1.25. Given symmetric monoidal category $\mathcal{V}$(use κ to denote unit), a *planar operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{C(j) \in \mathcal{V}\}_{j\in\mathbb{N}}$, a *unit map* $\eta : 1 \to C(1)$, together with composition maps

$$\gamma : C(k) \otimes C(j_1) \otimes ... \otimes C(j_k) \to C(j_1 + ... + j_k)$$

that satisfies the following axioms:

- *Unital Axiom:*

$$\begin{tikzcd} C(j) \otimes \kappa^{\otimes j} \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{C(j)}\otimes\eta^{\otimes j}"'] & & C(j) \\ & C(j) \otimes C(1)^{\otimes j} \arrow[ur, "\gamma"'] & \end{tikzcd}, \qquad \begin{tikzcd} \kappa \otimes C(j) \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_{C(j)}"'] & & C(j) \\ & C(1) \otimes C(j) \arrow[ur, "\gamma"'] & \end{tikzcd}$$

  These also guarantee the n-fold unit map $\eta^{\otimes j}$ is a monomorphism, and γ for involving only index j and 1 are epimorphisms.

- *Associativity Axiom:* Fix natural numbers k, $j = \sum_{s=1}^{k} j_s$, define index $g_s := j_1 + ... + j_s$ for $1 \leq s \leq k$. For each $i_1, ..., i_{g_k} = i_j$, with sum $i = \sum_{r=1}^{j} i_r$, and $h_s := i_{g_{(s-1)}+1} + ... + i_{g_s}$ (so $i = \sum_{s=1}^{k} h_s$ also), the following diagram commutes:

$$\begin{tikzcd} \left[C(k) \otimes \left(\bigotimes_{s=1}^{k} C(j_s)\right)\right] \otimes \left(\bigotimes_{r=1}^{j} C(i_r)\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & C(j) \otimes \left(\bigotimes_{r=1}^{j} C(i_r)\right) \arrow[dr, "\gamma"] & \\ & & C(i) \\ C(k) \otimes \left[\bigotimes_{s=1}^{k} \left(C(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} C(i_{g_{(s-1)}+q})\right)\right)\right] \arrow[r, "\mathrm{id}_{C(k)}\otimes(\bigotimes_s \gamma)"'] & C(k) \otimes \left(\bigotimes_{s=1}^{k} C(h_s)\right) \arrow[ur, "\gamma"'] & \end{tikzcd}$$

From my intuitive understanding, unital axiom is saying "action with unit $\eta : \kappa \to C(1)$" is doing nothing on any other objects $C(j)$, regardless of the order it's composed.

On the other hand, associativity axiom is accounting for associativity when "joining different actions": For instance, given $k = 3$, $j_1 = 2, j_2 = 2, j_3 = 3$ (so $j = 7$), and $i_\ell = \ell$ (so $h_1 = 3, h_2 = 7, h_3 = 18$, and $i = 28$). ne can first perform the action $C(3) \otimes (C(j_1 = 2) \otimes C(j_2 = 2) \otimes C(j_3 = 3)) \to C(j = 7)$, then perform the action $C(7) \otimes \left(\bigotimes_{\ell=1}^{7} C(i_\ell)\right) \to C(i = 28)$.

Alternatively, one can let each $C(j_s)$ act on $\bigotimes_{q=1}^{j_s} C(i_r g_{(s-1)} + q)$ (for instance, $C(j_1 = 2)$ action $C(j_1 = 2) \otimes C(i_1 = 1) \otimes C(i_2 = 2) \to C(3)$), which collects to a morphsim to $C(3) \otimes C(7) \otimes C(18)$; finally, consider the action of $C(k = 3)$ on these three.

The associativity is requiring any such pair of actions must commute, reducing the ambiguity of the action.

Remark 1.26. Most of the time, there is an extra requirement on the unit, namely $C(0) := \kappa$, and the action morphism $\gamma : C(0) \to C(0)$ (namely the action of nothing) is $\mathrm{id}_{C(0)}$.

Definition 1.27. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \to \mathcal{C}'$, is a sequence of morphisms $\psi_j : C(j) \to C'(j)$ in $\mathcal{V}$, such that it preserves the operad's structure morphism:

$$\begin{tikzcd} C(k) \otimes C(j_1) \otimes ... \otimes C(j_k) \arrow[r, "\gamma"] \arrow[d, "\psi_k \otimes \psi_{j_1} \otimes ... \otimes \psi_{j_k}"'] & C(j_1 + ... + j_k) \arrow[d, "\psi_{j_1+...+j_k}"] \\ C'(k) \otimes C'(j_1) \otimes .. \otimes C'(j_k) \arrow[r, "\gamma'"'] & C'(j_1 + ... + j_k) \end{tikzcd}$$

Example 1.28. (Endomorphism Operad on Based Spaces) First, recall that the category of based spaces $\mathcal{T}$ is a symmetric monoidal category under regular cartesian product $\times$, and it's self enriched (as all homset has induced topology from regular function space, and has a basepoint called constant basepoint map).

For each based space $X \in \mathcal{T}$ (with $*$ denotes the basepoint), consider the collection $\mathcal{E}_X(j) := F(X^j, X)$ (all based map from jth cartesian product $X^j \to X$). Notice that give $j = j_1 + ... + j_k$, given based maps $g_i \in \mathcal{E}_X(j_i)$, the universality of product guarantees k maps $g_i \circ \pi_i : \prod_{i=1}^k X^{j_i} = X^j \to X^{j_i} \to X$, hence induces a based map $\prod g_i : X^j \to X^k$. Define the action morphism as follow:

$$\gamma : \mathcal{E}_X(k) \times \prod_{i=1}^{k} \mathcal{E}_X(j_i) \to \mathcal{E}_X(j), \quad \gamma(f; g_1, ..., g_k) := f \circ (\prod g_i) : X^j \to X^k \to X$$

Which, $\mathcal{E}_X(0) = F(*, X)$ is a one-point set (as one can only include the basepoint to the basepoint of X), and the unit map $\eta : * \to \mathcal{E}_X(1)$ is $* \mapsto \mathrm{id}_X$. This forms an operad, after the routine check using the universality of product extensively...

This example of operad is encoding how "product of morphisms" can be communicated in a multi-entry sense (also talks about how such communication acts on X, in the most elementary way).

For operads in spectra (specifically about symmetric and orthogonal spectra), each space (1-pt compactification of some n-dimensional real inner product space) is associated with a nth symmetric group Σ_n action (once we fix a coordinate for instance).

Definition 1.29. An *operad* $\mathcal{C}$ in $\mathcal{V}$, is a planar operad, such tha each $C(j)$ is equipped with a right Σ_j-action. More formally, each $C(j)$ equips with a group homomorphism $\phi_j : \Sigma_j \to \mathrm{Aut}_{\mathcal{V}}(C(j))$, recognizing each permutation as a morphism, and act on $C(j)$ by precomposition.

Additionally, the symmetric group action satisfies the following axiom:

- Given $\sigma \in \Sigma_k$, $j_1 + ... + j_k = \Sigma_j$, let $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes "permutation of k-blocks of letters".

  Example 1.30. Given $j = 5 = 2 + 3$, the permutation $(1\ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5 \rangle \mapsto \langle 3, 4, 5; 1, 2 \rangle$ (namely changing the blocks' order, without changing the order in each block).

  Then, the following diagram commutes:

$$\begin{tikzcd} C(k) \otimes \left(\bigotimes_{r=1}^k C(j_r)\right) \arrow[r, "\sigma \otimes \text{ shuffle}_{\sigma^{-1}}"] \arrow[d, "\gamma"'] & C(k) \otimes \left(\bigotimes_{r=1}^k C(j_{\sigma(r)})\right) \arrow[d, "\gamma"] \\ C(j) \arrow[r, "\sigma(j_{\sigma(1)}, ..., j_{\sigma(k)})"'] & C(j) \end{tikzcd}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus ... \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_rth block in j (for instance, let τ_1 acts on $1, ..., j_1$; let τ_2 acts on $j_1 + 1, ..., j_1 + j_2$, etc.). Then, the following diagram commutes:

$$\begin{tikzcd}
C(k) \otimes \left(\bigotimes_{r=1}^k C(j_r)\right) \arrow[r, "\mathrm{id}_{C(k)} \otimes (\bigotimes_r \tau_r)"] \arrow[d, "\gamma"'] & C(k) \otimes \left(\bigotimes_{r=1}^k C(j_r)\right) \arrow[d, "\gamma"] \\
C(j) \arrow[r, "\tau_1 \oplus ... \oplus \tau_k"'] & C(j)
\end{tikzcd}$$

The first axiom is "permuting blocks of entries", while the second one is "permuting each block of entries", combining the two it provides about how to permute all finite amount of actions.

Example 1.31. Back to the endomorphism operad example, each $\mathcal{E}_X(j)$ has an obvious Σ_j-action, by permuting the j coordinates involved. Which, by definition when combining $\mathcal{E}_X(k) \times \left(\prod_{r=1}^k \mathcal{E}_X(j_r)\right) \to \mathcal{E}_X(j)$, we first combine the rth block $\{j_1 + ... + j_{r-1} + 1, ..., j_1 + ... + j_{r-1} + j_r\}$ into one entry, the Σ_k-action permutes the blocks of j_r; and, the internal permutation of j_r also carries over naturally.

This also suggests there is a natural way to let Σ_j acts on any object in the monoidal category: More precisely, each $\sigma \in \Sigma_j$ has a shuffle morphism $\mathrm{shuffle}_{\sigma^{-1}} X^{\otimes j} \to X^{\otimes j}$. By convention, it acts on the left (postcomposing).

Definition 1.32. Let be $\mathcal{C}$ an operad on $\mathcal{V}$. An object $A \in \mathcal{V}$ is a $\mathcal{C}$*-algebra* if it's equipped with a collection of morphisms $\theta : C(j) \otimes A^{\otimes j} \to A$ for $j \geq 0$, such that the following holds:

- *Associativity Axiom:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{tikzcd}
\left[C(k) \otimes \left(\bigotimes_{i=1}^k C(j_i)\right)\right] \otimes A^{\otimes j} \arrow[r, "\gamma \otimes \mathrm{id}_{A^{\otimes j}}"] \arrow[dd, "\text{shuffle}"'] & C(j) \otimes A^{\otimes j} \arrow[dr, "\theta_j"] & \\
& & A \\
C(k) \otimes \left[\bigotimes_{i=1}^k \left(C(j_i) \otimes A^{\otimes j_i}\right)\right] \arrow[r, "\mathrm{id}_{C(k)} \otimes \left(\bigotimes_i \theta_{j_i}\right)"'] & C(k) \otimes A^{\otimes k} \arrow[ur, "\theta_k"'] &
\end{tikzcd}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit Axiom:* The following diagram commutes:

$$\begin{tikzcd}
\kappa \otimes A \arrow[rr, "\sim"] \arrow[dr, "\eta \otimes \mathrm{id}_A"'] & & A \\
& C(1) \otimes A \arrow[ur, "\theta_1"'] &
\end{tikzcd}$$

- *Eauivariance Axiom:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{tikzcd}
C(j) \otimes A^{\otimes j} \arrow[rr, "\sigma \otimes \text{ shuffle}_{\sigma^{-1}}"] \arrow[dr, "\theta_j"'] & & C(j) \otimes A^{\otimes j} \arrow[dl, "\theta_j"] \\
& A &
\end{tikzcd}$$

Example 1.33. Notice the classical example in $\mathcal{T}$ the category of based spaces, the endomorphism operad $\mathcal{E}_X$ naturally acts on $X \in \mathcal{T}$, by defining $\theta_j : \mathcal{E}_X(j) \times X^j \to X$ as evaluation: $\theta_j(f; x_1, ..., x_n) := f(x_1, ..., x_n)$ (let's handwave the continuity part here :P).

The multilinearity of breaking up entries is compatible as long as the shuffle of X^j to A^{j_i} preserves order, the unit axiom is immediate when we define the "unit"in $\mathcal{E}_X(1)$ as id_X, and the equivariance axiom is satisfied because "shuffling the input" in $\mathcal{E}_X(j)$ and "shuffling the entries" in X^j have inverse meanings, so shuffling entries of X^j according to the inverse of $\sigma \in \Sigma_j$ will cancel the action on $\mathcal{E}_X(j)$, causing it to be identity.

More explicitly, the actions of σ are as follow, with the left on $\mathcal{E}_X(j)$ (precomposing the shuffle), the right on X^j (shuffle using the inverse):

$$f(x_1, ..., x_j) \mapsto f(x_{\sigma(1)}, ..., x_{\sigma(j)}), \qquad (x_1, ..., x_j) \mapsto (x_{\sigma^{-1}(1)}, ..., x_{\sigma^{-1}(j)})$$

Here, there is a more distinct reason than monad about why it's called "algebra", because we can let A act on other objects again:

Definition 1.34. Given $\mathcal{C}$ an operad on $\mathcal{V}$, $A \in \mathcal{V}$ a $\mathcal{C}$-algebra. An object $M \in \mathcal{V}$ is an *A-module*, if it's equipped with morphisms $\lambda_j : C(j) \otimes A^{\otimes(j-1)} \otimes M \to M$ for $j \geq 1$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{tikzcd}
\left[C(k) \otimes \left(\bigotimes_{i=1}^k C(j_i)\right)\right] \otimes A^{j-1} \otimes M \arrow[r, "\gamma\otimes\mathrm{id}_{A^{j-1}\otimes M}"] \arrow[dd, "\text{shuffle}"'] & C(j) \otimes A^{j-1} \otimes M \arrow[dr, "\lambda_j"] & \\
 & & M \\
C(k) \otimes \left[\bigotimes_{i=1}^{k-1} \left(C(j_i) \otimes A^{j_i}\right)\right] \otimes \left(C(j_k) \otimes A^{j_k-1} \otimes M\right) \arrow[r, "\mathrm{id}_{C(k)}\otimes(\bigotimes_i \theta_{j_i})\otimes\lambda"'] & C(k) \otimes A^{k-1} \otimes M \arrow[ur, "\lambda_k"'] &
\end{tikzcd}$$

- *Unit:* Special case when $j = 1$ (so contains $A^0 = \kappa$), the following diagram commutes:

$$\begin{tikzcd}
\kappa \otimes M \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_M"'] & & M \\
 & C(1) \otimes M \arrow[ur, "\lambda"'] &
\end{tikzcd}$$

- *Equivariance:* Given $\sigma \in \Sigma_{j-1} \hookrightarrow \Sigma_j$ (in the canonical way $\{1, ..., j-1\} \hookrightarrow \{1, ..., j-1, j\}$), the following diagram commutes:

$$\begin{tikzcd}
C(j) \otimes A^{j-1} \otimes M \arrow[rr, "\sigma\otimes\,\text{shuffle}_{\sigma^{-1}}\otimes\mathrm{id}_M"] \arrow[dr, "\lambda_j"'] & & C(j) \otimes A^{j-1} \otimes M \arrow[dl, "\lambda_j"] \\
 & M &
\end{tikzcd}$$

Right now I can only think of a trivial example...where every $\mathcal{C}$-algebra A has itself being an A-module.

However, I did see that if the symmetric monoidal category $\mathcal{V}$ has an *internal hom functor* (that is, a bifunctor $\mathcal{H} : \mathcal{V}^{\mathrm{op}} \times \mathcal{V} \to \mathcal{V}$, such that a natural isomorphism $\mathrm{Hom}(A \otimes B, C) \cong \mathrm{Hom}(A, \mathcal{H}(B, C))$ exists). In this case, $(\_ \otimes A \dashv \mathcal{H}(A, \_))$ is an adjoint pair, and $\mathcal{V}$ is called *closed symmetric monoidal category*. Using this $\mathcal{H}$ to define the endomorphism operad, then all the properties carry over.

Some of the variants of operads include:

1. *Planar Operads*, this often happens when we want to omit the commutativity structure of certain algebra. Professor May also defined this to be *non-*Σ *operad*.

2. *Unital Operads*, these are the special case where $C(0) = \kappa$, insisting the "0-ary operation" (or the operation of "doing nothing") agrees with the traditional sense. However, if the algebra we want to work on doesn't have units (for instance, Lie algebras as analogy in traditional algebra), then $C(0) := 0$ (the initial object of the category) would be more sensible.

3. *Augmentation Operads*, these are unital operads with *augmentation morphisms* $\epsilon := \gamma : C(j) \otimes C(0)^j \to C(0) = \kappa$, and also the *degeneracy maps* as follow:

$$\sigma_{ji} : C(j) \cong C(j) \otimes C(0)^j \xrightarrow{\mathrm{id}_{C(j)} \otimes \eta^{i-1} \otimes \mathrm{id}_{C(0)} \otimes \eta^{j-1}} \left[C(j) \otimes C(1)^{i-1} \otimes C(0) \otimes C(1)^{j-i}\right] \xrightarrow{\gamma} C(j-1)$$

   where $1 \leq i \leq j$.

Some other examples:

Example 1.35. There is a trivial operad, defined by $\mathcal{N}(j) := \kappa$, with each $\gamma : \mathcal{N}(k) \otimes \mathcal{N}(j_1) \otimes ... \otimes \mathcal{N}(j_k) \to \mathcal{N}(j)$ being the canonical isomorphism of products of κ. Notice the isomorphism by definition respects permutation (i.e. all permutation yields the same isomorphism).

An object $A \in \mathcal{V}$ that's an $\mathcal{N}$-a;gebra (with morphisms $\theta_j : \mathcal{N}(j) \otimes A^j \to A$) can be regarded as a monoid, because we have a unit morphism $\theta_1 : \mathcal{N}(1) \otimes A \cong \kappa \otimes A \xrightarrow{\sim} A$ by definition, and together with associativity diagram as follow:

$$\begin{tikzcd}
\mathcal{N}(2) \otimes \mathcal{N}(1) \otimes \mathcal{N}(2) \otimes A^3 \arrow[r, "\gamma \otimes \mathrm{id}_{A^3}"] \arrow[d, "\text{shuffle}"'] & \mathcal{N}(3) \otimes A^3 \arrow[dr, "\theta_3"] & \\
& & A \\
\mathcal{N}(2) \otimes (\mathcal{N}(1) \otimes A) \otimes \left(\mathcal{N}(2) \otimes A^2\right) \arrow[r, "\mathrm{id}_{\mathcal{N}(2)} \otimes \theta_1 \otimes \theta_2"'] & \mathcal{N}(2) \otimes A^2 \arrow[ur, "\theta_2"'] &
\end{tikzcd}$$

If switch the order of $\mathcal{N}(1)$ and $\mathcal{N}(2)$, this demonstrates the agreement with the other way of composition, hence the "action" $\mathcal{N}(j) \otimes A^j \to A$ disregards the order of "composition", which is associative. Finally, with the permuation action being trivial, this states the multiplication action is commutative, hence A can be regarded as a commutative monoid.

(If disregard the permutation action, A becomes just a monoid).

Similarly, $M \in \mathcal{V}$ is an A-module, if it's the classical A-module by chasing through similar monoidal actions.

This example is also called *associative non-*Σ *operad*, because it's a non-Σ operad that produces associative monoids as its algebra. For further details check the presentation notes this week :)